CLOUD COMPUTING SECURITY

MECR1043

SECTION 52

LECTURER

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LAB 2

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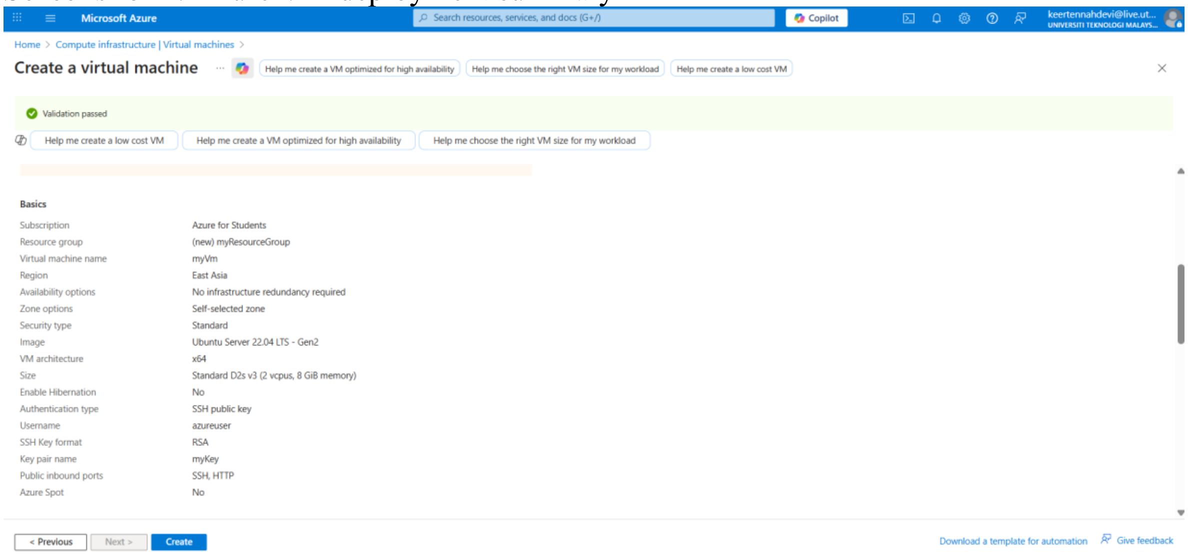
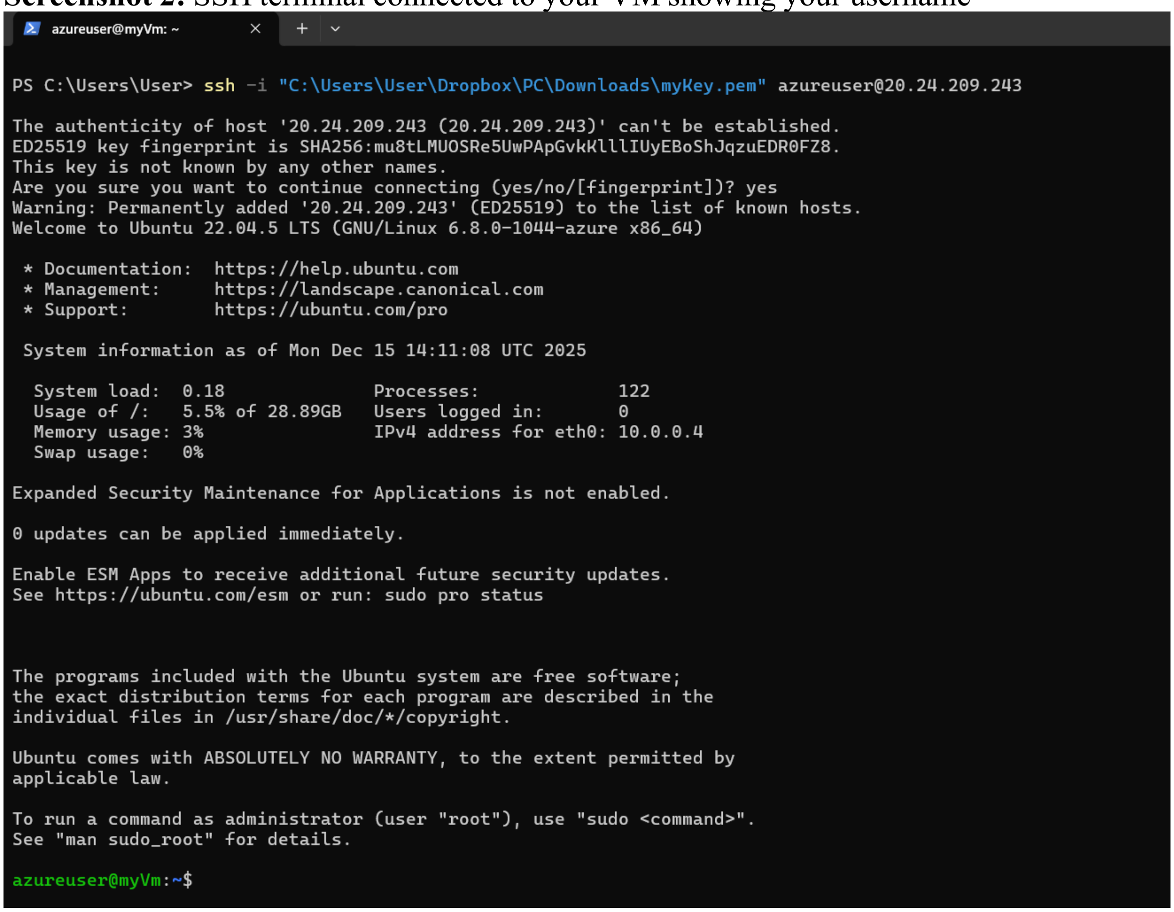
Screenshot 1: Azure VM deployment summary



Screenshot 2: SSH terminal connected to your VM showing your username



Screenshot 3: Output of uname -a from inside the VM

Reflection(Skillset)

During this lab, I successfully acquired and reinforced several key technical skills essential for working with cloud environments.

I learned how to navigate the Azure portal to provision a new VM, which involved specifying core parameters such as the subscription, creating a new resource group (myResourceGroup), naming the VM (myVM), and selecting the appropriate image (Ubuntu Server 22.04 LTS - Gen2). This task built foundational competency in using a major cloud provider's web-based interface for infrastructure setup.

Next, I implemented the industry-standard SSH public key authentication method, generating a new key pair (myKey), and safely downloading the private key file (myKey.pem). Crucially, I utilized this key pair to establish a secure, remote connection to the Linux VM from a local machine using the SSH client, demonstrating hands-on experience in cloud network security and access.

Another than that, I practiced configuring basic network security by defining Inbound port rules, specifically enabling access for SSH (port 22) and HTTP (port 80). This demonstrated an understanding of how to manage firewall-like rules to control public internet access to the VM.

Post-connection, I executed fundamental Linux commands inside VM’s shell. This included using uname -a to verify the VM's kernel and architecture, and commands like sudo apt-get -y update and sudo apt-get -y install nginx to manage packages and install a web server. This validated a basic level of proficiency in managing a Linux system via the command line.

Finally, the lab culminated in the installation of the NGINX web server and the verification of its successful deployment by accessing VM’s public IP address through a web browser, which confirms the ability to deploy and validate application-level services.

Conclusion, these skills are directly relevant to roles involving DevOps, cloud engineering, and system administration, providing a tangible starting point in the domain of Cloud Computing Security